UNIT 6 · TOPIC 7

Wheel Slip and Sinkage

Syllabus topic: Slippage and sinkage of wheels

Wheel slip (p. 245; ASAE S296.2)

$$\text{Slip (travel reduction) } S = 1 - V_a/V_t,\ V_t = r\omega$$

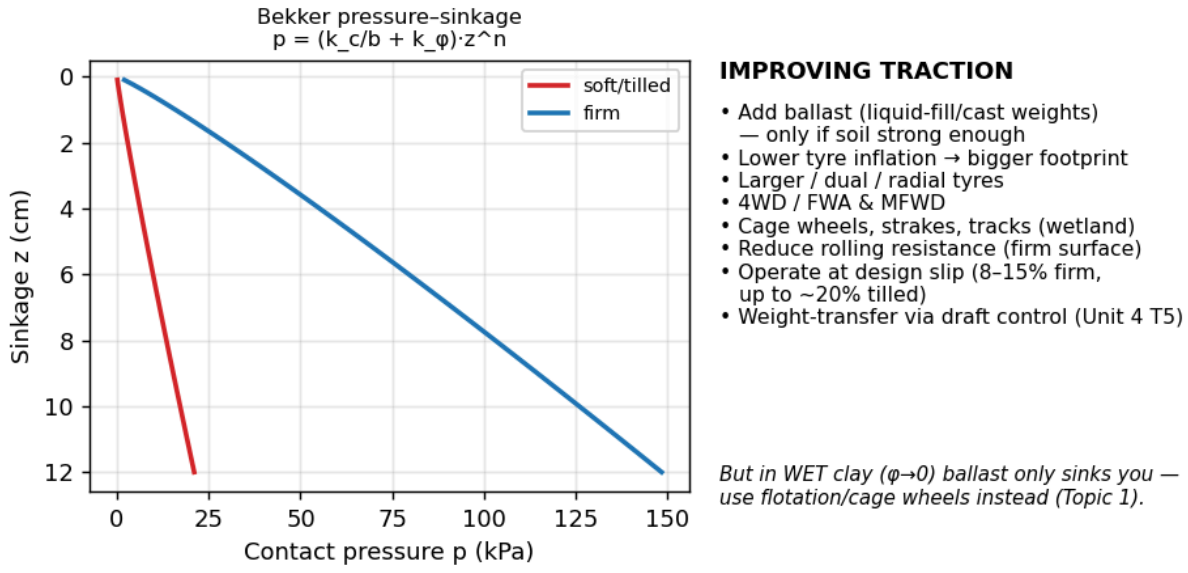
V_a = actual travel speed; V_t = theoretical speed; r = rolling radius on a hard surface, defined per **ASAE S296.2**.

- A driving wheel MUST slip to develop thrust — thrust rises with slip toward a maximum.
- Measure slip by comparing wheel revolutions over a distance, loaded vs unloaded (Unit 3, Topic 2b).
- **Typical working slip: 8–15%** on firm soil, up to ~**20%** on tilled soil.
- Excessive slip wastes energy and shears/pulverises the surface.

Sinkage — Bekker pressure–sinkage

$$p = (k_c/b + k_\varphi)\cdot z^n$$

z = sinkage under contact pressure p; k_c = cohesive and k_φ = frictional sinkage moduli; n = sinkage exponent; b = smaller contact dimension. The parameters come from a **BEVAMETER** (plate-sinkage) test.



Bekker pressure–sinkage curves (soft/tilled vs firm) and ways to improve traction

- Sinkage causes compaction resistance — a major part of rolling resistance. If sinkage exceeds what the tyre can manage, traction collapses.
- **Reduce sinkage with lower ground pressure:** larger/wider tyres, lower inflation, tracks, flotation devices.

Practice MCQs

Q77. Wheel slip is defined as S =:

(a) V_t/V_a (b) $1 - V_a/V_t$
(c) $V_a - V_t$ (d) $V_a/V_t + 1$

Q78. In the slip formula, V_t is the:

(a) Actual travel speed (b) Theoretical speed rω on a hard surface
(c) Wind speed (d) PTO speed

Q79. A driving wheel develops thrust only when it:

(a) Is perfectly rigid (b) Slips (some travel reduction)
(c) Stops (d) Is towed

Q80. Rolling radius is standardised in:

(a) ASAE S296.2 (b) ASAE S313.2
(c) ISO 500 (d) SAE J833

Q81. Typical working slip on firm soil is about:

(a) 0% (b) 8–15%
(c) 40% (d) 80%

Q99. Sinkage under contact pressure follows Bekker's:

(a) $p = (k_c/b + k_\varphi)\cdot z^n$ (b) $p = z/b$
(c) $p = c + \sigma\tan\varphi$ (d) $p = \pi DN/60$

Q100. Pressure–sinkage parameters are obtained with a:

(a) Cone penetrometer only (b) Bevameter (plate-sinkage)
(c) Patternator (d) Dynamometer

Answers

Q77 — (b) $1 - V_a/V_t$ — *p. 245 — $S = 1 - V_a/V_t$.*
Q78 — (b) Theoretical speed rω on a hard surface — *$V_t = r\omega$, rolling radius per ASAE S296.2.*
Q79 — (b) Slips (some travel reduction) — *A wheel must slip to develop thrust.*
Q80 — (a) ASAE S296.2
Q81 — (b) 8–15%
Q99 — (a) $p = (k_c/b + k_\varphi)\cdot z^n$
Q100 — (b) Bevameter (plate-sinkage)

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10, pp. 240–252; Wismer & Luth (1974), Off-road traction prediction for wheeled vehicles, Trans. ASAE; Freitag (1965) mobility-number dimensional analysis; Bekker pressure–sinkage theory; ASAE S296.2 (slip/rolling radius), S313.2 (cone index); NPTEL traction lectures.